
Numerical Simulation of the Wake of a Confined Obstacle

Bénard–von Kármán Instability

Determination of the Critical Reynolds Number using the Finite Element Method

Téo ROCCAMATISI, BENJAMIN ÖZTÜRK

Fluid Mechanics 2026

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1 Physical Introduction

The wake of a bluff body (circular or square cross-section) exhibits, for Reynolds numbers on the order of a few tens, an instability known as the **Bénard–von Kármán** instability: beyond a critical threshold Re_c , the steady and symmetric flow gives way to a street of alternating vortices, which are periodic in time.

This numerical study aims to:

1. Determine the instability threshold Re_c using a bisection method.
2. Characterize the growth of the oscillations via the exponential envelope $e^{\sigma_r t}$ and measure the growth rate σ_r .
3. Verify the Stuart–Landau scaling law: $|A| \propto (\text{Re} - \text{Re}_c)^{1/2}$.

The software `FreeFem++` is used, solving the incompressible Navier–Stokes equations using the finite element method.

2 Incompressible Navier–Stokes Equations

2.1 Strong Formulation

We seek the velocity field $\mathbf{u}(\mathbf{x}, t) = (u_x, u_y)$ and the pressure field $p(\mathbf{x}, t)$ satisfying, within the domain Ω :

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \eta \Delta \mathbf{u} \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

By non-dimensionalizing lengths by the obstacle width d , velocities by the flow velocity U , times by d/U , and pressures by ρU^2 , these equations become:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}, \quad \nu = \frac{1}{\text{Re}} \quad (3)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (4)$$

where the Reynolds number is $\text{Re} = Ud/\nu_{\text{kin}}$.

2.2 Boundary Conditions

- **Inlet** (boundary d): parabolic Poiseuille profile $u_x = \frac{6y(B-y)}{B^2}$, $u_y = 0$, with a mean velocity $\langle u_x \rangle = 1$.
- **Channel walls** (boundaries a, c) and **obstacle** (boundary $e-h$): no-slip condition $\mathbf{u} = \mathbf{0}$.
- **Outlet** (boundary b): free outlet condition, $u_y = 0$, zero pressure gradient.

3 Determination of the Critical Threshold by Bisection

Experimentally, the Reynolds number found by the Solène - Clément team is **50**.

Simulation Parameters:

Blockage ratio (B/d)	: 10
Channel length	: 100
Mesh resolution	: 1
Refinement on the obstacle	: 3
Automatic mesh adaptation	: yes
Time step Δt	: 0.1
Number of iterations	: 1000
Measurement point position x (rel. obstacle)	: 0
Measurement point position y (rel. obstacle)	: 5

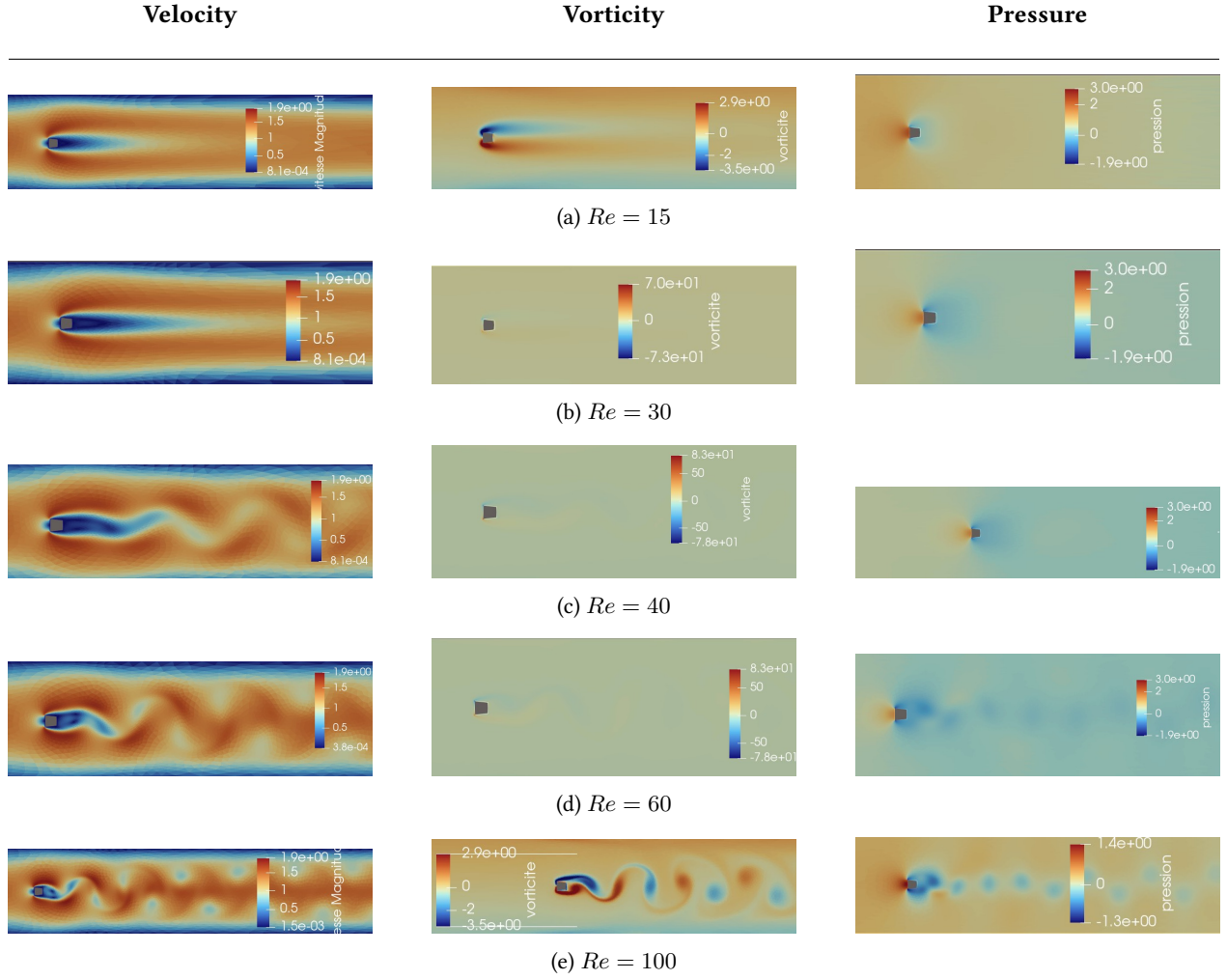


Figure 1: Velocity, vorticity, and pressure fields for different Reynolds numbers Re .

For $Re < 30$ (**subcritical regime**), the wake is perfectly symmetrical with respect to the centerline of the channel. Two steady recirculation zones are attached to the rear of the obstacle; they elongate slightly as Re increases but remain stable. The vorticity field displays two symmetrical lobes of opposite signs, without downstream propagation. The pressure map shows a strong gradient at the front of the obstacle (stagnation overpressure) and a low-pressure region downstream confined to the recirculation zone. These features are typical of the extended Stokes regime ($Re \ll Re_c$), where viscous forces dominate and stabilize any perturbation.

At $Re > 40$ (**supercritical regime**), the von Kármán instabilities appear. Alternating vortices periodically shed from each side of the obstacle.

4 Exponential Envelope and Growth Rate

4.1 Linear Regime

In the vicinity of the threshold, for sufficiently small amplitudes, the dynamics are governed by the linear part of the Stuart–Landau equation. Denoting $A(t)$ as the amplitude of the oscillations (e.g., the envelope of $u_y(t)$), we have in the linear regime:

$$\frac{dA}{dt} = \sigma_r A \quad \implies \quad A(t) = A_0 e^{\sigma_r t} \quad (5)$$

The complete oscillations are then written as:

$$u_y(t) = A_0 e^{\sigma_r t} \cos(\sigma_i t + \varphi_0) \quad (6)$$

where $\sigma_i = 2\pi f_1$ is the natural angular frequency of the instability and φ_0 is an initial phase shift.

4.2 By Bisection

The critical Reynolds number Re_c is the threshold below which the flow returns to the steady symmetric state after any perturbation, and beyond which the oscillations are self-sustained.

It is determined using a bisection method. This yields a Reynolds number of **33**.

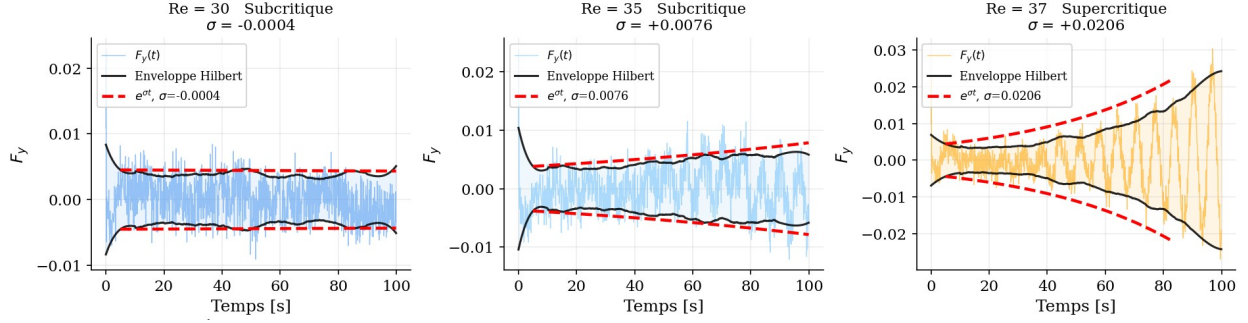


Figure 2: Extraction of σ_r alongside the other parameters characterizing the signal.

Using a Python routine, we can extract the growth rate σ_r by fitting the Hilbert envelope of the signal $u_y(t)$.

- $Re = 30$, $\sigma_r = -0.0004$. The exponential envelope decays slowly: $\sigma_r < 0$ confirms that $Re = 30 < Re_c$ and that the flow is subcritical. The decay is very weak (almost zero value), indicating that this Reynolds number is close to, but below, the critical threshold.
- $Re = 35$, $\sigma_r = +0.0076$. The envelope grows weakly but distinctly: $\sigma_r > 0$ indicates a supercritical regime. The value remains small, placing this Reynolds number slightly above the threshold.

4.3 Intersection of Lines

We can also plot the lines $\sigma = f(Re)$ for $Re > 0$ and for $Re < 0$. The intersection of these lines gives us Re_c .

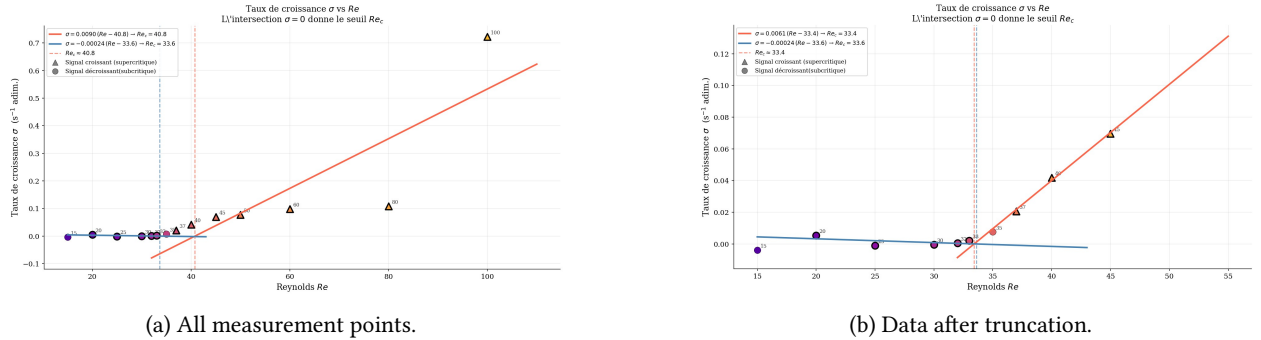


Figure 3: Comparison of the evolution of σ_r as a function of Re : full measurement points (a) and data truncated to the linear regime (b). The interpolation line crosses the $\sigma_r = 0$ axis at $Re_c \approx 33.4$.

- **Fig 3.(a) – full data.** When using all measurement points, the relation $\sigma_r(Re)$ is not linear for $\sigma_r > 0$: the curve becomes concave.
- **Fig 3.(b) – data truncated to the linear regime.** By keeping only the measurements taken when the amplitudes are still small enough for the dynamics to be governed by the linear part of the Stuart–Landau equation, we obtain a linear dependence, in agreement with the equation. The slope of the regression line provides the coefficient k of the law $\sigma_r = k(Re - Re_c)$, and the intersection with $\sigma_r = 0$ yields $Re_c \approx$ **33.4**.

5 Amplitude Law and Hopf Bifurcation

5.1 Stuart–Landau Equation

The Stuart–Landau equation describes the non-linear dynamics of the complex amplitude $A(t) \in \mathbb{C}$:

$$\frac{dA}{dt} = \sigma A - \frac{1}{2} l |A|^2 A \quad (7)$$

where $\sigma = \sigma_r + i\sigma_i$ is the complex growth rate and $l = l_r + il_i$ is the non-linear coefficient. Separating the real and imaginary parts and continuing the derivations, we can demonstrate the scaling law:

$$|A|_\infty \propto (\text{Re} - \text{Re}_c)^{1/2} \quad (8)$$

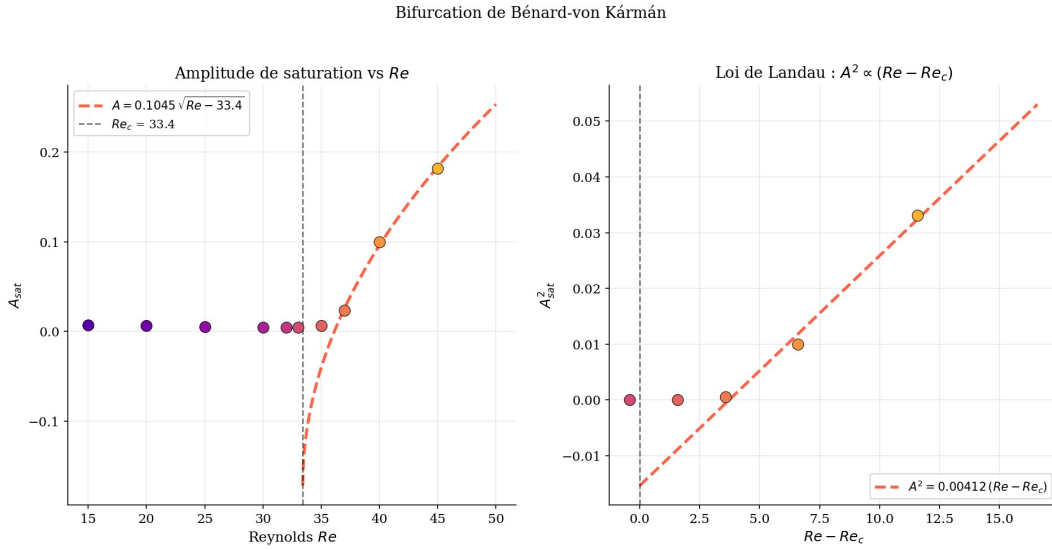


Figure 4: Bifurcation diagram of the Bénard–von Kármán instability. *Left*: saturation amplitude $|A|_\infty$ as a function of Re ; the curve follows $|A|_\infty \propto (\text{Re} - \text{Re}_c)^{1/2}$ with $\text{Re}_c \approx 33.4$. *Right*: representation of $|A|^2$ as a function of $(\text{Re} - \text{Re}_c)$ confirming the expected linear behavior.

6 Conclusion

This numerical study characterized the Bénard–von Kármán instability in a confined flow by solving the Navier–Stokes equations using finite elements in FreeFem++.

We first determined the critical Reynolds number using two complementary approaches. The bisection method, based on the qualitative observation of the signal, led to $\text{Re}_c \approx 33$, while the quantitative analysis of the growth rate σ_r and its linear extrapolation provided a more precise value of $\text{Re}_c \approx 33.4$.

The study of the weakly supercritical regime also revealed an exponential growth of perturbations, followed by non-linear saturation, confirming the amplitude law $|A| \propto (\text{Re} - \text{Re}_c)^{1/2}$.

A comparison with the experimental value $\text{Re}_c^{\text{exp}} \approx 50$ reveals a significant discrepancy. This is primarily explained by confinement effects (blockage ratio $B/d = 10$), which promote the instability and lower the critical threshold. Other factors may also play a role, such as the exact geometry of the obstacle, the two-dimensional nature of the simulation, as well as numerical choices (mesh resolution, boundary conditions, and time discretization).

References

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